1. W1: canonical Pic^0 arithmetic on smooth plane quartics

Run record 2026-10-18, branch `claude/quartic-pic-canonical-20261018`, frozen protocol commit `3fad80fa2`. Synthetic instances only; not compared with any external implementation (no Magma, Sage, PARI or msolve on the host), so no fastest-implementation claim, and no key-recovery, sub-rho or deployed-curve claim. Every number below is **measured** by native Rust unless marked derived. Raw records: `results/`.

1.1. Question and answer

Can $\mathrm{Pic}^0(C)(\mathbb{F}_q)$ of a smooth plane quartic be given arithmetic with class-unique keys using only $\mathbb{F}_q$ linear algebra, so that the degree-6 key collisions of `quartic_ic.rs` disappear, and what does one operation cost as q grows? Answer (measured): yes. Keys are the echelon matrix of the conics through the unique effective degree-3 divisor of $x + D_\infty$ (typical classes) or the point P of the class $K - P$ (exceptional classes). All validation tests pass (V1 to V5, below). A generic addition costs 3523 multiplications and 44 inversions for every q from 2^{12} to 2^{31} (3503 at $q = 251$), about 4343 multiplication-equivalents at $q = 2^{31} - 1$.

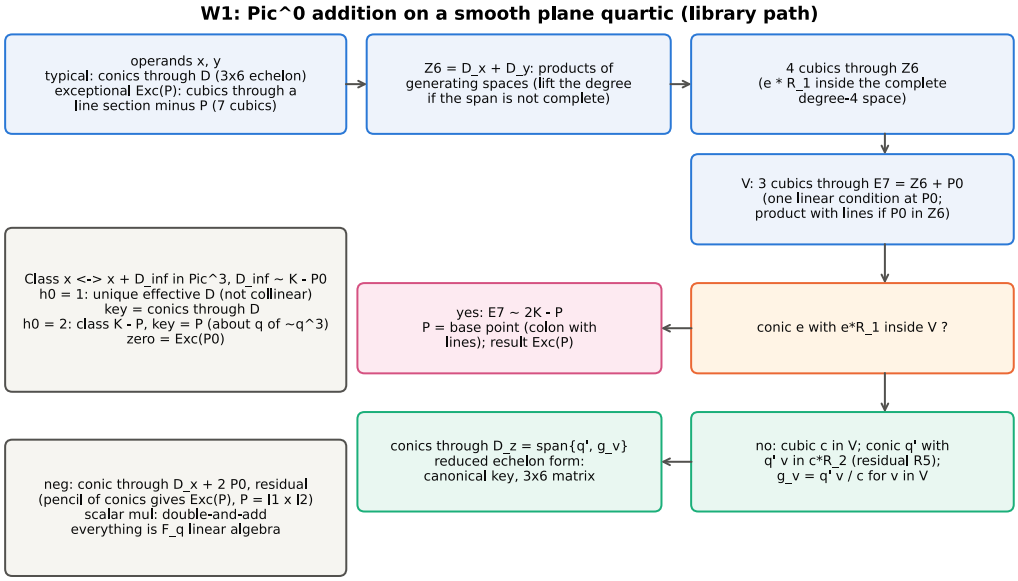


Figure 1: The addition pipeline (diagram of the algorithm, `reports/diagram.py`).

1.2. Validation (exact pass/fail)

V1	21 instances (nine GHS quartics, three random each at q = 5, 7, 11, 13): enumerated degree-3 divisors equal the zeta coefficient; distinct keys = #Jac from the L-polynomial (148, 356, 1348, 1324 on GHS; 1410, 1892, 3173 at q = 13); exceptional keys = #C(F_q).
V2	300 random triples per instance (associativity, commutativity, inverse, representative independence, auxiliary-cubic independence), #Jac · x = 0 for 100 x, full Cayley tables at q = 5 (87 and 148 classes, 658 503 and 3 241 792 triples), 14 650 exceptional pairs, 6 141 agreements with an independent second reduction path.
V3	key equality iff equal divisors or collinear with the same residual point, on every pair of enumerated divisors (23 016 same-residual pairs).
V4	GHS transfers at q = 5, 7, 11 (table below): homomorphism, order l, scalar relation hold on every pair; no unexplained canonical collision.
V5	smoothness certificate accepts all instances, rejects node, cusp, two conics, quadruple line.

q	l	usable n	deg-6 keys	old deg-3 keys	canonical keys	pairs	hom. fail	l x != 0	x(n) != n x(1)	unexplained	old collisions resolved
5	37	18	18	17	18	66	0	0	0	0	1
7	89	38	38	32	38	236	0	0	0	0	7
11	331	164	164	144	164	6536	0	0	0	0	20

V4 uses the split instance `quartic_ic.rs` builds at each q (records 1, 4, 8 of the GHS run); the other six GHS quartics are covered by V1 to V3, V5. Skipped by the dumper: 6, 6, 0 transfers (non-reduced degree-6 divisors or cubic space dimension not 4).

1.3. Operation cost (M1, M2)

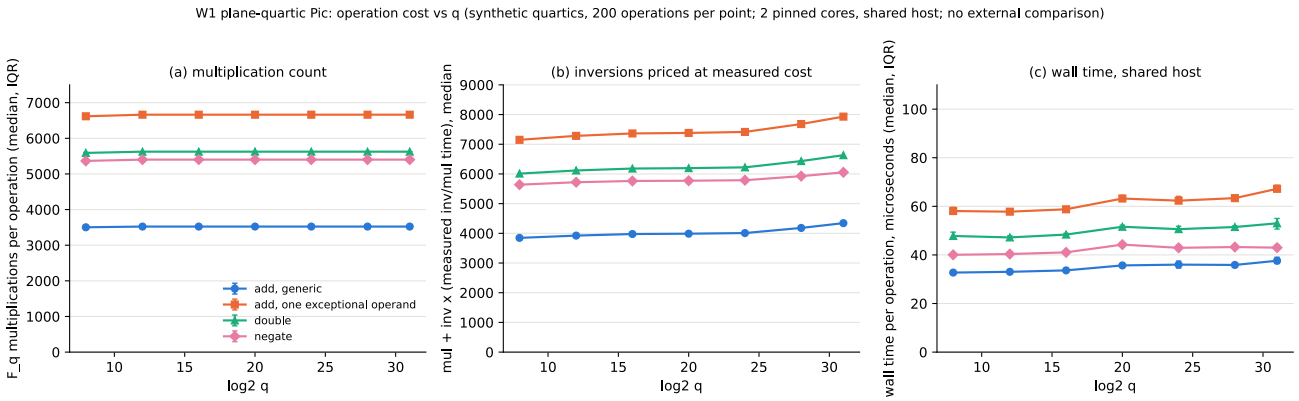


Figure 2: Median and interquartile range over 200 operations per point, scaling quartics from `results/m1_summary.jsonl`. (a) $\mathbb{F}_q$ multiplications; (b) with inversions priced at the measured inv/mul time ratio (7.8 at q = 251, 18.6 at q = 2^31 - 1); (c) wall time on two pinned cores of a host with load average above 80.

operation	mul	inv	wall at 2^8 (μs)	wall at 2^31 (μs)	priced at 2^31
add, generic	3523	44	32.7	37.6	4343
add, exceptional operand	6663	68	58.1	67.2	7930
double	5625	54	47.8	53.0	6631
negate	5402	35	40.1	43.0	6054
key encoding	36	3	0.7	0.8	92
scalar mul, 64-bit	467 111	4810	4190	4373	556 748

Fits against $\log_2 q$ (7 points, 95% interval): generic add multiplications $0.56[-0.25, 1.37]$ per bit (flat), priced $18.5[9.5, 27.4]$ per bit, wall $207[136, 278]$ ns per bit. Of 1400 generic adds none lifted a generating space or took an exceptional branch. Over the small-q tests 927 of 132 994 operations lifted, 16 945 had P_0 in the support, 21 126 ended exceptionally.

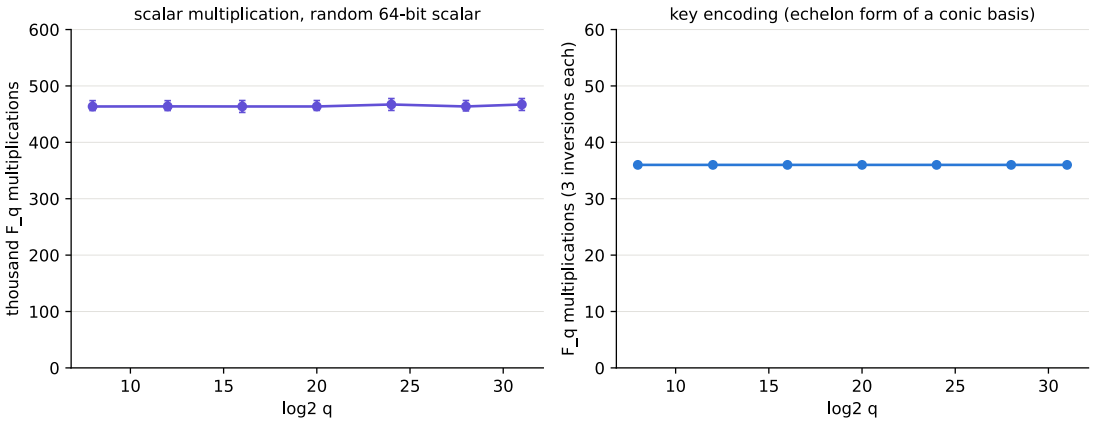


Figure 3: Scalar multiplication by a random 64-bit scalar and key encoding, same records.

1.4. Limits and deviations

The constants are those of this dense-linear-algebra implementation; they are counts, not bounds. V4 covers three of the nine GHS quartics (limited by reusing `quartic_ic.rs` unmodified). The base shift of the degree-6 transfer is inferred from the data and then verified on every pair. M1 and the tests ran through a stand-alone crate including the module; the pinned repository `cargo test --lib` result is in the PR. Details: README.md.